

Projection Operation in Relational Algebra and SQL

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1. Introduction

The **Projection** operation in **Relational Algebra (RA)** is used to **select specific columns (attributes)** from a relation. It removes unwanted attributes and can also remove duplicate tuples (since RA treats relations as sets).

In **SQL**, projection is achieved by listing only the required columns in the **SELECT** statement.

Key Points

- Projection focuses on **columns**, unlike Selection which focuses on **rows**.
- In RA, duplicates are automatically removed. In SQL, we must explicitly use **DISTINCT**.
- Attribute order does not matter in RA, but output column order in SQL follows the query.

2. Syntax

Relational Algebra

$$\pi_{A_1, A_2, \dots, A_n}(R)$$

Where $A_1, A_2, \dots, A_n$ are attribute names, and R is a relation.

SQL

```
SELECT column1, column2, ...  
FROM table_name;
```

3. Examples

Example 1: Basic Projection

Relation:

Employee(EmpID, Name, Dept, Salary)

Goal: Retrieve only Name and Dept.

Relational Algebra:

$\pi_{\text{Name,Dept}}(\text{Employee})$

SQL:

```
SELECT Name, Dept
FROM Employee;
```

Example 2: Removing Duplicates

If multiple employees have the same Name and Dept:

Relational Algebra:

$\pi_{\text{Name,Dept}}(\text{Employee})$

Automatically removes duplicates.

SQL:

```
SELECT DISTINCT Name, Dept
FROM Employee;
```

Example 3: Projection with Selection

Retrieve Name and Salary for employees in the IT department.

Relational Algebra:

$\pi_{\text{Name,Salary}}(\sigma_{\text{Dept}='IT'}(\text{Employee}))$

SQL:

```
SELECT Name, Salary
FROM Employee
WHERE Dept = 'IT';
```

Example 4: Projection with Computed Columns

Calculate the annual salary of all employees.

Relational Algebra: Extended projection is used (not in basic RA):

$\pi_{\text{Name, Salary} \times 12 \rightarrow \text{AnnualSalary}}(\text{Employee})$

SQL:

```
SELECT Name,
       Salary * 12 AS AnnualSalary
FROM Employee;
```

Example 5: Projection on Multiple Tables (Join + Projection)

List employee names with their department location.

Relational Algebra:

$$\pi_{\text{Name, Location}}(\text{Employee} \bowtie_{\text{Employee.Dept=Department.DeptName}} \text{Department})$$

SQL:

```
SELECT E.Name,  
       D.Location  
FROM Employee E  
JOIN Department D  
  ON E.Dept = D.DeptName;
```

4. Properties of Projection in RA

- **Idempotent:**

$$\pi_{A_1, A_2}(\pi_{A_1, A_2, A_3}(R)) = \pi_{A_1, A_2}(R)$$

- **Order Irrelevant:**

$$\pi_{A, B}(R) = \pi_{B, A}(R)$$

- Removes duplicates automatically (SQL requires DISTINCT).

5. Difference: Selection vs Projection

Aspect	Selection	Projection
Focus	Rows (tuples)	Columns (attributes)
RA Notation	$\sigma_{\text{condition}}(R)$	$\pi_{\text{attr list}}(R)$
SQL Equivalent	WHERE clause	SELECT column list
Duplicates	Preserves duplicates	Removes duplicates in RA

6. Summary

Projection is a core operation in both Relational Algebra and SQL, used to limit output to specific attributes. When combined with selection or joins, it becomes a powerful tool for querying exactly the data needed.